

Why constancy on the geometric generic fiber kills inertia

A Lamport-style expansion of the key step in Lemma 69

The sentence being expanded

The proof says:

Hence this restriction is constant, so its inertia at the generic point of the hyperplane at infinity is trivial. Unramifiedness can be checked after this separable extension of the residue field; consequently $\mathcal{P}^{[d]}$ is unramified at the generic point of H .

The purpose of this note is to make every implication in that sentence explicit. The main point is that two different facts are being used:

- (i) the triviality of the *tame* fundamental group makes the torsor constant on the geometric generic open fiber; and
- (ii) extending the residue field of a henselian discrete valuation ring by a separable extension does not erase ramification.

Notation and the local question

Put

$$X = \tilde{C}_{\mathfrak{m}}^{(d)}, \quad S = \mathrm{Pic}_{C, \mathfrak{m}}^d, \quad U = U^{(d)}, \quad D = H = X \setminus U.$$

Let $f: X \rightarrow S$ be the compactified Abel–Jacobi morphism. Let ξ be the generic point of S , and let η be the generic point of D . On the generic fiber one has

$$(X_\xi, D_\xi) \simeq (\mathbb{P}_{k(\xi)}^r, \mathbb{P}_{k(\xi)}^{r-1}), \quad U_\xi \simeq \mathbb{A}_{k(\xi)}^r.$$

Thus η is also the generic point of the hyperplane D_ξ .

Let $G = \Lambda^\times$. The G -torsor $\mathcal{P}^{[d]}$ on U gives a finite étale cover of the fraction field of the local ring at η . The local question is:

Does this cover extend étale across the divisor D ?

Structured proof

<1>1. Reduce unramifiedness to an inertia calculation.

Let

$$R = \mathcal{O}_{X, \eta}^h, \quad K = \mathrm{Frac}(R).$$

Because X is regular and D has codimension one, R is a henselian discrete valuation ring. Choose a geometric point of $\mathrm{Spec} K$ and let I_η be the corresponding inertia group.

Proof. A finite étale K -scheme extends to a finite étale R -scheme if and only if I_η acts trivially on its geometric fiber. Applied to $\mathcal{P}^{[d]}$, this gives the exact target:

$$\mathcal{P}^{[d]} \text{ is unramified at } \eta \iff I_\eta \text{ acts trivially on } \mathcal{P}_K^{[d]}.$$

⟨1⟩2. Pass to a separably closed generic field.

Choose a separable closure

$$\Omega = k(\xi)^{\text{sep}}.$$

It is enough for the étale argument that Ω be separably closed; it need not be an algebraic closure. After base change to $\text{Spec } \Omega$,

$$(X_\Omega, D_\Omega, U_\Omega) \simeq (\mathbb{P}_\Omega^r, \mathbb{P}_\Omega^{r-1}, \mathbb{A}_\Omega^r).$$

Let $\bar{\eta}$ be the generic point of D_Ω .

⟨2⟩1. The point $\bar{\eta}$ lies above η .

Proof. The divisor D is the relative hyperplane in the projective-space bundle $X \rightarrow S$. Hence its generic point remains the generic point of the hyperplane after passing from the generic field $k(\xi)$ to Ω .

⟨2⟩2. Locally, this base change only enlarges the residue field.

Let R_Ω denote the henselian discrete valuation ring at $\bar{\eta}$. The map $R \rightarrow R_\Omega$ sends a local equation of D to a local equation of D_Ω . In particular, its ramification index is one. Its residue-field extension is

$$\kappa(\eta) \longrightarrow \kappa(\bar{\eta}),$$

obtained by adjoining the separable constants $\Omega/k(\xi)$. Consequently, R_Ω/R is an unramified extension of henselian discrete valuation rings, possibly after taking a filtered union when Ω is infinite.

⟨2⟩3. The inertia group does not change.

Proof. After choosing compatible separable closures, an unramified extension of henselian discrete valuation fields identifies their geometric inertia groups:

$$I_{\bar{\eta}} \xrightarrow{\sim} I_\eta.$$

Equivalently, a nontrivial ramification index cannot become 1 merely by separably enlarging the residue field. Thus unramifiedness may be *checked* after this base change.

⟨1⟩3. Use the previously proved tameness.

The torsor $\mathcal{P}^{[d]}$ is assumed tame along D . Therefore its restriction to $U_\Omega = \mathbb{A}_\Omega^r$ is tame along $D_\Omega = \mathbb{P}_\Omega^{r-1}$.

Proof. In local terms, tameness says that the wild inertia subgroup $P_{\bar{\eta}} \subset I_{\bar{\eta}}$ already acts trivially. The only possible remaining local monodromy is the prime-to-characteristic quotient

$$I_{\bar{\eta}}^t = I_{\bar{\eta}}/P_{\bar{\eta}}.$$

This is why tameness alone does *not* yet prove unramifiedness.

⟨1⟩4. The tame fundamental group forces the restricted torsor to be constant.

The result cited in Lemma 69 gives

$$\pi_1^t(\mathbb{A}_\Omega^r; \mathbb{P}_\Omega^{r-1}) = 1.$$

Proof. Finite covers of $\mathbb{A}_\Omega^r$ that are tame along the hyperplane at infinity are classified by finite quotients of this tame fundamental group. Since the group is trivial, there is no nontrivial connected tame cover. Hence

$$\mathcal{P}^{[d]}|_{\mathbb{A}_\Omega^r} \simeq \mathbb{A}_\Omega^r \times G$$

as a G -torsor, after choosing one point in a geometric fiber. This is the meaning of “the restriction is constant.” The trivialization is not canonical, but its existence is all that is needed.

⟨1⟩5. A constant torsor has trivial inertia at infinity.

Proof. The constant torsor on $\mathbb{A}_\Omega^r$ extends as the constant finite étale torsor

$$\mathbb{P}_\Omega^r \times G \longrightarrow \mathbb{P}_\Omega^r.$$

It is therefore étale at $\bar{\eta}$, the generic point of the hyperplane at infinity. By Step ⟨1⟩1, this is equivalent to triviality of the action of $I_{\bar{\eta}}$.

One can say the same thing directly in terms of the two parts of inertia:

$$\begin{aligned} P_{\bar{\eta}} &\text{ acts trivially because the torsor is tame,} \\ I_{\bar{\eta}}^t &\text{ acts trivially because the global tame monodromy is trivial.} \end{aligned}$$

Hence the full group $I_{\bar{\eta}}$ acts trivially.

⟨1⟩6. Descend the conclusion to the original generic point.

Proof. By Step ⟨2⟩3, the map $I_{\bar{\eta}} \rightarrow I_\eta$ identifies the relevant inertia actions. Since the action becomes trivial after the separable residue-field extension, it was already trivial before that extension. Step ⟨1⟩1 now gives

$\mathcal{P}^{[d]} \text{ is unramified at the generic point } \eta \text{ of } H.$

What the original three sentences compress

The logic is the following chain:

$$\begin{array}{c}
\mathcal{P}^{[d]} \text{ is tame along } H \\
\Downarrow \\
\text{its pullback to } \mathbb{A}_\Omega^r \text{ is classified by } \pi_1^t(\mathbb{A}_\Omega^r; \mathbb{P}_\Omega^{r-1}) \\
\Downarrow \quad (\pi_1^t = 1) \\
\text{the pulled-back torsor is constant} \\
\Downarrow \\
I_{\bar{\eta}} \text{ acts trivially} \\
\Downarrow \quad (I_{\bar{\eta}} \simeq I_\eta) \\
I_\eta \text{ acts trivially} \\
\Downarrow \\
\mathcal{P}^{[d]} \text{ is unramified at } \eta.
\end{array}$$

The indispensable input is the *global* statement that the relative tame fundamental group of the affine fiber is trivial. Without that input, tameness would kill only wild inertia and could leave nontrivial tame inertia.

A thesis-ready replacement paragraph

Here is a more explicit replacement for the quoted passage:

Let $\Omega = k(\xi)^{\text{sep}}$, and let $\bar{\eta}$ be the generic point of the hyperplane $H_\Omega \simeq \mathbb{P}_\Omega^{r-1}$. The pullback of $\mathcal{P}^{[d]}$ to $\mathbb{A}_\Omega^r$ is tame along H_Ω . Since

$$\pi_1^t(\mathbb{A}_\Omega^r; \mathbb{P}_\Omega^{r-1}) = 1,$$

this pullback is a constant $\Lambda^\times$ -torsor. It therefore extends as the constant finite étale torsor over $\mathbb{P}_\Omega^r$, so the inertia group $I_{\bar{\eta}}$ acts trivially. The passage from the generic point η of H to $\bar{\eta}$ is an unramified extension of the corresponding henselian discrete valuation rings: it enlarges the residue field separably and leaves a uniformizer unchanged. Hence $I_{\bar{\eta}} \simeq I_\eta$, and triviality of inertia can be checked after this base change. Thus I_η acts trivially, which says precisely that $\mathcal{P}^{[d]}$ is unramified at the generic point of H .

A small notational caution

If $\overline{k(\xi)}$ denotes an algebraic closure, then in positive characteristic the extension $\overline{k(\xi)}/k(\xi)$ need not be separable. For the argument above, one may first use the separable closure $k(\xi)^{\text{sep}}$. Passing afterwards through a purely inseparable extension does not change finite étale covers or their inertia. This makes the phrase “after this separable extension of the residue field” literally correct.